

Package / Proxy Server: General Settings / General

General Remote Cache Local Cache Antivirus ACLs Traffic Mgmt Authentication Users Real Time Sync

Squid General Settings

Enable Squid Proxy ☒ Check to enable the Squid proxy.
Note: If unchecked, ALL Squid services will be disabled and stopped.

Keep Settings/Data ☒ If enabled, the settings, logs, cache, A/V defs and other data will be preserved across package reinstalls.
Note: If disabled, all settings and data will be wiped on package uninstall/reinstall/upgrade.

Proxy Interface(s)
OPT1
WAN
[Dropdown]
The interface(s) the proxy server will bind to.
Note: Use CTRL + click to select multiple interfaces.

Proxy Port
This is the port the proxy server will listen on.
(Default: 3128)

ICP Port
This is the port the proxy server will send and receive ICP queries to and from neighbor caches.
Leave this blank if you don't want the proxy server to communicate with neighbor caches through ICP.

Allow Users on Interface ☒ If checked, the users connected to the interface(s) selected in the 'Proxy Interface(s)' field will be allowed to use the proxy.
There will be no need to add the interface's subnet to the list of allowed subnets.

Patch Captive Portal ☐ This feature was removed - see bug #5584 for details!
If you were using this feature, double check /etc/inc/captiveportal.inc/ content for sanity.
Get a sane copy of the file from pfSense GitHub repository if needed.

Reverse DNS IPv4 First ☐ Enable this to force DNS IPv4 lookup first. This option is very useful if you have problems accessing HTTPS sites.

Disable ICMP ☐ Check this to disable Squid ICMP ping/pong.

Use Alternate DNS Servers for the Proxy Server ☐ If you want to use DNS servers other than the DNS forwarder/resolver configured in pfSense, enter the IP(s) here.
Note: Separate entries by semi-colons (,)

Transparent Proxy Settings

Transparent HTTP Proxy ☒ Enable transparent mode to forward all requests for destination port 80 to the proxy server without any additional configuration being necessary.
Note: Transparent mode will filter SSL (port 443) if you enable man-in-the-middle options below.
In order to proxy both HTTP and HTTPS protocols without interrupting SSL connections, configure WPAD/PAC options on your DNS/DHCP servers.

Transparent Proxy Interface(s)
OPT1
WAN
The interface(s) the proxy server will transparently intercept requests on.
Note: Use CTRL + click to select multiple interfaces.

Bypass Proxy for Private Address Destination ☐ Do not forward traffic to Private Address Space (RFC 1918) destinations.
Destinations in Private Address Space (RFC 1918) are passed directly through the firewall, not through the proxy server.

Bypass Proxy for These Source IPs ☐ Do not forward traffic from these source IPs, CIDR nets, hostnames, or aliases through the proxy server but let it pass directly through the firewall. (Applies only to transparent mode.)
Note: Separate entries by semi-colons (,)

Bypass Proxy for These Destination IPs ☐ Do not proxy traffic going to these destination IPs, CIDR nets, hostnames, or aliases, but let it pass directly through the firewall. (Applies only to transparent mode.)
Note: Separate entries by semi-colons (,)

Logging Settings

Enable Access Logging ☒ This will enable the access log.
Warning: Do not enable if disk capacity is low.

Log Store Directory
The directory where the logs will be stored. This is also used for logs other than the Access Log above.
Default: /var/squid/logs
Note: Do NOT include the trailing / when setting a custom location.

Rotate Logs
Defines how many days of logfiles will be kept. Rotation is disabled if left empty.

Figure 5: Squid proxy configuration

Basic pfSense installation is really simple and straightforward. Boot the system using the CD created from the ISO image and answer very simple questions. Care should be taken at one step – select 'I' to start installation on the hard disk when prompted to select from

'Recovery' and 'Installer' mode. Once this process is complete, reboot the system.

Earlier versions of pfSense had live boot CD support, which is no more available. But the live boot functionality is possible using a USB drive.

The next step is to assign interfaces. For this, select the *Assign Interfaces* menu option at the prompt. All the Ethernet interfaces recognised by pfSense, along with their MAC IDs and their link status, will be visible. Select the appropriate interfaces for your LAN, WAN and OPT1 connections. Figure 1 depicts the set-up used for this test installation.

From the menu, proceed to 'Set Interface IP' and assign the LAN interface IP address. Once that is done, the box is ready to be controlled from the Web based interface. Connect to pfSense's Web interface from the system connected to the Ethernet switch. This system should be configured with the IP address in the range of the pfSense LAN IP.

System default login name is admin and password is pfSense. As a best practice, make sure to change your password from the *System – User manager – Actions* menu immediately after the first log-on.

Configuration

pfSense supports various WAN protocols such as DHCP, Static IP, PPP, PPTP, PPPoE, etc. In this test set-up, a PPPoE WAN connection has been used. Under *Interfaces – WAN – PPPoE connection*, fill in the user name, password, service name (optional) and save the configuration. Go through Figures 2 and 3 for detailed settings. The relevant explanation about these settings is available on the page. It is also possible to configure pfSense using *System – Setup Wizard* menu.

Assuming the cable from the PPPoE modem to the pfSense WAN port already exists and that the modem/Internet connection is in working condition, the Internet will connect readily. Use the logs available under *Status – System Logs* to troubleshoot if you face any problems.

An interesting feature called Dial-On-Demand (DoD) is available for the PPPoE connection. If there is no traffic towards the Internet for 'Idle timeout' seconds, the DoD mode will disconnect WAN, and on the next Internet request, the WAN will reconnect.

The basic pfSense LAN – WAN setup is now complete. It will help you if you browse through the pfSense Web based menu for getting acquainted with it, before continuing with